

# ■ NEUROLOGICAL AI DIAGNOSTIC IMAGING REPORT

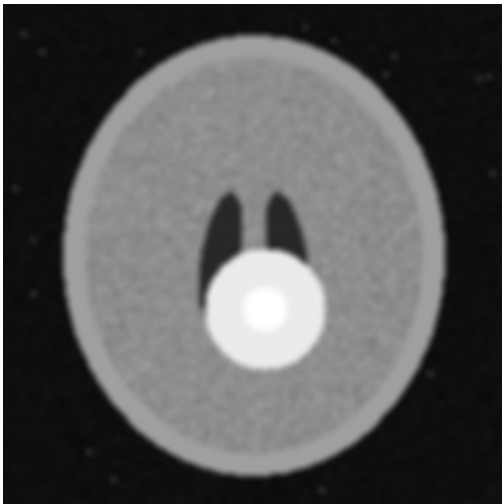
Automated Deep Learning & Grad-CAM Explainability Analysis

|              |                         |               |                                   |
|--------------|-------------------------|---------------|-----------------------------------|
| Case ID:     | SAMPLE-REPORT-001       | Modality:     | Brain MRI (Axial / Coronal)       |
| Date / Time: | 2026-08-28 16:13:23 UTC | Model Engine: | ResNet50 Fine-Tuned + Bayesian MC |

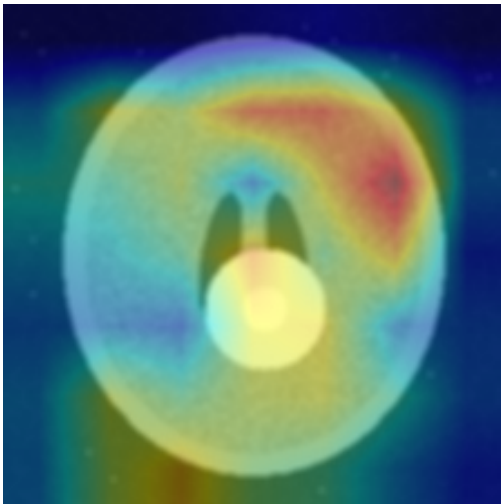
## 1. Primary Diagnostic Findings

|                           |                              |
|---------------------------|------------------------------|
| Predicted Classification: | MENINGIOMA                   |
| Model Confidence:         | 42.36%                       |
| Epistemic Uncertainty:    | 0.0570 (Entropy Index: 0.94) |
| Clinical Review Status:   | HIGH_RISK_RADIOLOGIST_REVIEW |

## 2. Visual Explainability (Grad-CAM Attention Mapping)



Original MRI Scan



Grad-CAM Attention Heatmap

## 3. Multi-Class Probability & Variance Distribution

| Class Name | Probability (%) | Epistemic Std (±%) | Category        |
|------------|-----------------|--------------------|-----------------|
| Glioma     | 25.15%          | ±6.22%             | Tumor Neoplasm  |
| Meningioma | 42.36%          | ±7.62%             | Tumor Neoplasm  |
| Pituitary  | 13.53%          | ±4.65%             | Tumor Neoplasm  |
| No Tumor   | 18.96%          | ±4.33%             | Healthy Control |

■■ **REGULATORY & CLINICAL DISCLAIMER:** This diagnostic report was generated automatically by an experimental AI research pipeline. It has NOT been cleared by regulatory bodies (e.g. FDA / CE) and must NOT be used as the sole basis for clinical diagnosis or treatment. Final evaluation requires certified radiologist sign-off.